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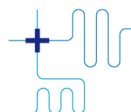
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# TIME-TEMPERATURE SUPERPOSITION APPLIED TO PBX MECHANICAL PROPERTIES

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**Abstract.** The use of plastic-bonded explosives (PBXs) in weapon applications requires that they possess and maintain a level of structural/mechanical integrity. Uniaxial tension and compression experiments are typically used to characterize the mechanical response of materials over a wide range of temperatures and strain rates, providing the basis for predictive modeling in more complex geometries. After many years of data collection on a variety of PBX formulations, we have here applied the principles of time-temperature superposition to a mechanical properties database which includes PBX 9501, PBX 9502, PBXN-110, PBXN-9, and HPP (propellant). Consistencies are demonstrated between the results of quasi-static tension and compression, dynamic Split-Hopkinson Pressure Bar (SHPB) compression, and cantilever Dynamic Mechanical Analysis (DMA). Time-temperature relationships of maximum stress and corresponding strain values are analyzed, in addition to the more conventional analysis of modulus. The extensive analysis shows adherence to the principles of time-temperature superposition and correlations of mechanical response to binder glass-transition temperature ( $T_g$ ) and specimen density. Direct ties exist between the time-temperature analysis and the underlying basis of a useful existing PBX mechanical model (ViscoSCRAM). Results give confidence that, with some limitations, mechanical response can be predicted at conditions not explicitly measured.

**Keywords:** PBX, PBXN, mechanical properties, time-temperature analysis.

**PACS:** 62.20.-x, 65.40.De.

## INTRODUCTION

The thermo-mechanical response of plastic-bonded explosives (PBX) determines their resistance to various physical stimuli and insults. Mechanical characterization allows one to trend response as a function of specific parameters, to detail failure criteria in the material, and to calibrate models that may ultimately predict mechanically-induced ignition. After years of obtaining an experimental database that includes various test configurations, conditions and PBX materials, we here provide a universal and useful analysis using time-temperature principles. This approach has found previous success in

applications to polymers [1] and PBX compression data [2]. Our analysis, broadly applied to compression, tension, and dynamic oscillation, indicates various levels of consistency in response and provides a basis for the interpolation and limited extrapolation of PBX mechanical response beyond what is actually measured.

## EXPERIMENTAL PROCEDURE

The PBX materials used in this study are PBX 9501 and PBX 9502, PBXN-9, PBXN-110 and HPP (high performance propellant). Most of the specimens used to obtain the data presented here were machined out of large isostatically-pressed

charges. Specimens for quasi-static compression are typically 9.525 mm diameter by 19.05 mm long, and tensile “dogbones” are cylindrical with a gauge length of 38.1 mm and 12.7 mm diameter. Specimens for SHPB are typically 6 mm diameter by 3 mm long, and DMA bars are 17 x 12.7 x 3.2 mm. Quasi-static tests were performed on an Instron 5568 with a Bemco environmental chamber for temperature control; two oppositely-mounted knife-edge extensometers are used to measure the average strain as described previously [3]. The SHPB apparatus/technique has been described in detail elsewhere [4]. The DMA measurements were performed on a TA Instruments Model 2980 [5]. In the case of HPP tension, due to the softness of the material, tensile strain was measured by Cheng Liu using digital image correlation techniques [6].

Stress-strain results from constant strain rate quasi-static tests to failure were reduced to three representative parameters for this analysis. For each test, the maximum stress value (Max Stress) was determined, along with its corresponding strain (Max Strain). Also, the Modulus (slope of stress vs strain) at 25% of the maximum stress was calculated. For SHPB, the rising edge of the stress-strain curve is not in stress equilibrium and only the stress maximum term can be determined. Between 2 and 5 specimens were tested at each temperature and strain rate combination, the averaged parameters used in the time-temperature analysis shown here.

## TIME-TEMPERATURE BACKGROUND

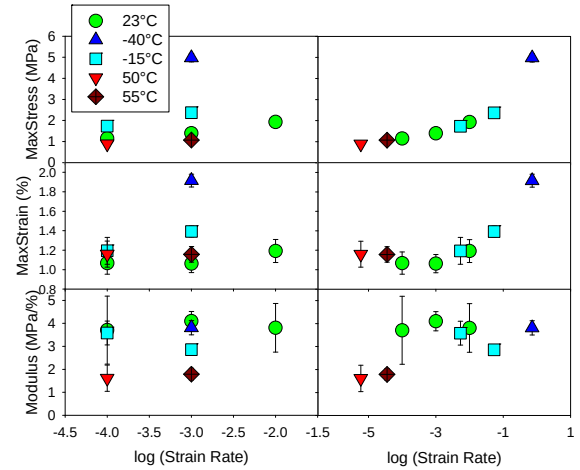
In the earliest reference to time-temperature analysis, Glasstone et al. [7] relates differences in frequency response of a material to temperature changes via an activation barrier. The well-known and widely-used Williams-Landel-Ferry (WLF) model [8] relates viscosity and temperature using shift factors. In 2005, Siviour et al. [1] applied a simplified time-temperature relationship to polymeric data, and in 2008, Williamson [2] successfully applied it to PBX compressive strength data:

$$\log a_T = \log \left( \frac{\dot{\epsilon}_{ref}}{\dot{\epsilon}} \right) = \frac{1}{A} (T_{ref} - T) \quad (1)$$

This expression directly relates all changes in temperature to factor-of-ten changes in strain rate with a single coefficient,  $A$ , for a given material. While the different historical time-temperature equations give somewhat similar results, we are attracted to the simplicity of Eq. 1. Our objective has been to expand the application of this equation to include stress, modulus and ultimate strain data, as well as tensile and DMA data, from a large variety of PBX materials.

## RESULTS AND DATA ANALYSIS

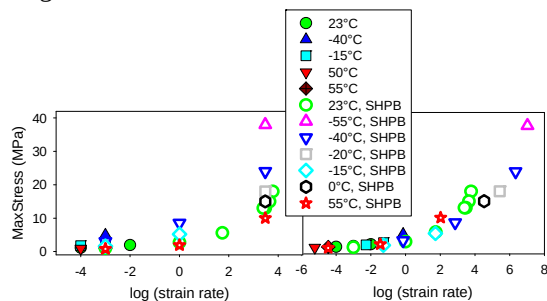
Much of the PBX 9501 and 9502 data, with or without the time-temperature discussion, has been published previously [2-5]. In Fig. 1a are plotted values of the three test parameters from PBXN-9 quasi-static compression tests at various temperatures and strain rate. Using a single number,  $A = -22^\circ \text{C} / \text{decade}$  in Eq. 1, the maximum stress, strain and modulus values were shifted to an “equivalent  $23^\circ \text{C}$  temperature” on the strain rate axis to produce the “master curve” shown in Fig. 1b.



**Figure 1.** Quasi-static compression parameters for PBXN-9, (a) left, as-measured maximum stress (top), strain (center) and modulus (bottom); (b) right, master curve, same parameters shifted to equivalent  $23^\circ \text{C}$ .

In Fig. 2a, the maximum stress parameters from quasi-static tests (Fig. 1a, top) are re-plotted with those from SHPB tests at higher rates. Again using  $A = -22^\circ \text{C} / \text{decade}$ , the stress values are

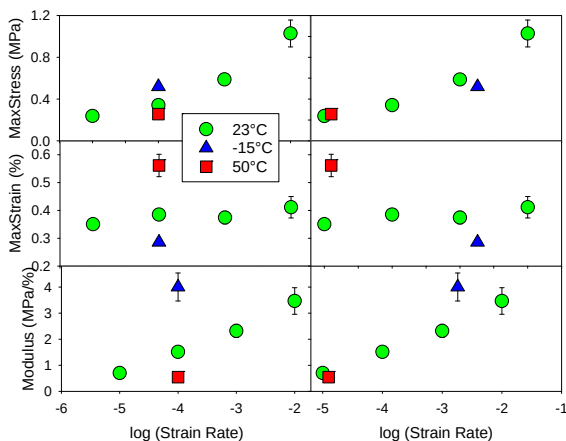
shifted to form the master curve in Fig. 2b. Note that this curve spans more than 10 orders of magnitude of strain rate.



**Figure 2.** Quasi-static and SHPB compressive stress for PBXN-9, (a) left, as-measured stress; (b) right, master curve, stress parameters shifted to equivalent 23° C.

Tensile parameters for PBXN-9 are shown in Fig. 3, before (a) and after (b) time-temperature shift using Eq. 1 and  $A = -30^\circ \text{C/decade}$ .

The time-temperature shifted DMA data is shown in Fig. 4 where  $A = -6^\circ \text{C/decade}$  was used. The raw, unshifted  $\log(\text{frequency})$  data are not shown, but vary between -1 and 1 (frequencies between 0.1 and 10 Hz).

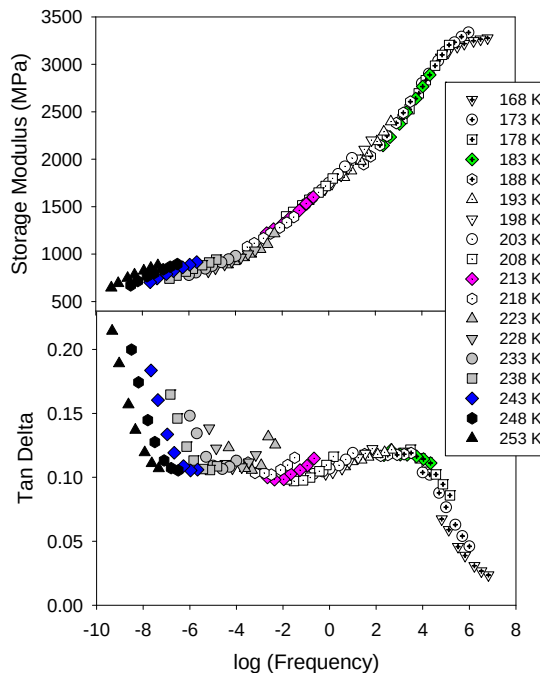


**Figure 3.** Quasi-static tension parameters for PBXN-9, (a) left, as-measured maximum stress (top), strain (center) and modulus (bottom); (b) right, master curve, same parameters shifted to equivalent 23° C.

## DISCUSSION

The time-temperature analysis of PBXN-9 data shown in Figs. 1 through 4 was repeated for

the other PBX materials listed in the Experimental Section with quite similar results. Fitting parameters are summarized in Table 1. In compression and tension, the maximum stress and modulus parameters shift to form satisfactory master curves for all PBX types *except* PBX 9501, where SHPB stress values are higher than what quasi-static time-temperature would predict [3]. Strain values in both tension and compression are a mixed bag, sometimes forming a satisfactory or less than satisfactory master curve.



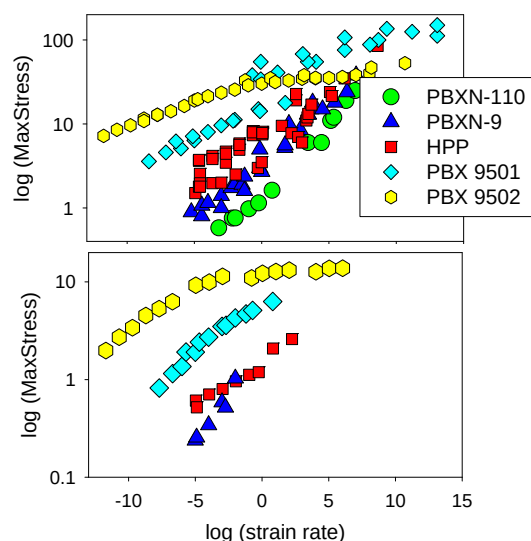
**Figure 4.** Shifted DMA data for PBXN-9, (a) top, storage modulus (MPa) and (b) bottom, tan delta.

DMA data for the PBXs adhere to time-temperature principles over large temperature ranges (note disruption of tan delta master curve for temperatures above 213 K,  $-60^\circ \text{C}$ ). However, as a rule, the shift factor  $A$  in Eq. 1 is smaller for DMA than for the quasi-static and SHPB data (for PBXN-9, the parameter  $A$  is -6 and -22, respectively). The DMA temperature range showing a satisfactory master curve varies for each PBX and the point of departure correlates roughly with the  $T_g$  of the binder. Figure 5 shows a summary of compressive and tensile master curves (equivalent 23° C) for all five PBX materials. In

these curves, the by-eye slopes correlate with the magnitude of the shift parameter A in Eq. 1. In compression, the by-eye master curve slopes correlate with PBX specimen density and possibly with binder  $T_g$  values (Table 1). Tensile master curves cover a smaller range of strain rates, but again relative position of the master curves (and magnitude of A) correlates with density and/or  $T_g$ .

**TABLE 1.** PBX physical properties and A shift factors (A\* is for DMA data, c is compression, t is tension).

|          | Density<br>(g/cm <sup>3</sup> ) | $T_g$<br>(°C) | A<br>(°C/dec)     | A*<br>(°C/dec) |
|----------|---------------------------------|---------------|-------------------|----------------|
| PBX 9502 | 1.890                           | 28            | -8                | -6             |
| PBX 9501 | 1.835                           | -45           | -8(c),<br>-10(t)  | -8             |
| HPP      | 1.820                           | -37           | -14               | na             |
| PBXN-9   | 1.752                           | -40           | -22(c),<br>-30(t) | -6             |
| PBXN-110 | 1.659                           | -89           | -22(c)            | na             |



**Figure 5.** Summary of PBX master curves (equivalent 23° C) for (a) top, quasi-static and SHPB compression, and for (b) bottom, quasi-static tension. Shift parameters using Eq(1) are listed in Table 1.

## CONCLUSIONS

Adherence to time-temperature principles does not insure the ability to interpolate or extrapolate

correct mechanical response, however, it provides some confidence in efforts to model trends and make predictions when limited data are available. In addition, corroboration of trends in material response by multiple tests (quasi-static and SHPB) is important and time-temperature analysis provides a framework for comparison. Time-temperature principles apply equally to tension and compression, and allow the relative response of a variety of PBXs to be considered together.

## ACKNOWLEDGEMENTS

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